

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've spent 10 years building backend systems and APIs, with the last 3 focused on architecture and technical leadership at AMA. Right now I own 4 REST APIs that serve 5 internal teams and external partners and process \$4.3M in annual transactions. I architected an event-driven serverless system on AWS that has held 99.9% uptime since launch and processed 2.8 million events. I built a self-service developer platform that took environment provisioning from 2-3 weeks down to 2 hours. Earlier in my career I built backend services for a medical imaging AI platform at Arterys. My day-to-day is Python, Java, and TypeScript with Postgres and Kubernetes. I lead through RFCs, code reviews, mentorship, and a seat on AMA's Architecture Steering Committee.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Own and maintain 4 critical REST APIs serving 5 internal teams and external partners. Designed API contracts, approve all new endpoint additions, and maintain API design standards documented in Confluence. APIs process \$4.3M in annual transactions. (Python, REST, OpenAPI, TypeScript)
- Architected technical infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, \$4.3M annual revenue, 2.8M insured trips processed, 99.9% uptime since launch. Won 2024 Guidewire Innovation Award. (Python, AWS Lambda, Terraform, TypeScript)
- Built self-service developer platform solo that reduced environment provisioning from 2-3 weeks to 2 hours (60x improvement). Adopted by 15 personnel across 3 teams, created 120+ environments. Unblocked major initiatives by enabling rapid iteration and production debugging via snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Architected modernization of legacy document generation system into containerized microservices on EKS. 4-5 services (with sidecars) on a 3-5 node cluster, deployed via Helm. Datadog observability for failure count, success count, generation time, and storage time. Leading 2-3 engineers on the project. (Kubernetes, EKS, Docker, Helm, Datadog, C#)
- Set technical standards for 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Member of Architecture Steering Committee. Help newer developers navigate legacy codebases and trained developers on TDD methodology.

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Java, Gosu)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Backend Architecture & APIs	Design Python services on AWS that run in production. Own and maintain 4 critical REST APIs with OpenAPI design standards, serving 5 internal teams and external partners and processing \$4.3M in annual transactions. Architected an event-driven serverless system that has held 99.9% uptime since launch and processed 2.8 million events.
Healthcare Backend	Built and maintained backend services that ran medical AI algorithms and returned results to the platform at Arterys. Improved AI algorithm processing turnaround using Redis caching and AWS infrastructure optimizations. Monitored production via Elasticsearch and Kibana, catching and resolving issues before customer impact.
Cloud & Infrastructure	Production AWS across multiple accounts (Lambda, EKS, SQS, RDS, IAM) with Terraform for infrastructure as code. Built and maintain a developer platform (60x faster provisioning), and a production EKS cluster running cert-manager, Datadog, Karpenter, and Linkerd.
Technical Leadership & Domain Ownership	Set technical standards for 12-person team through code reviews, RFC authorship, PR checklists, and API design guides. Member of AMA's Architecture Steering Committee. Drive cross-team architectural decisions including error handling and sanitization standards. Own technical domains from architecture through production.

Technologies

Backend Languages	Python, TypeScript, Java, Node.js, Bash, C#
Architecture & APIs	REST APIs, OpenAPI, Microservices, Event-Driven Architecture, Async Workflows
Databases	PostgreSQL, Aurora/RDS, Redis
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM), Terraform, Kubernetes, Docker, Helm
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, Alerting